Chapter 5: Special Probability Distributions

STAT2920: Introduction to Probability

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Outline

- There are a handful of discrete distributions which come up frequently in applications.
- Our topic is to study some of these special distributions and establish some of their properties.

Definition 1.

A random variable X has a discrete uniform distribution and it is referred to as a discrete uniform random variable if and only if its probability distribution is given by

$$f(x) = \frac{1}{k} \quad \text{for } x = x_1, x_2, \ldots, x_k$$

where $x_i \neq x_j$ when $i \neq j$

Remark 1.

In the special case where $x_i = i$, the discrete uniform distribution becomes

$$f(x) = \frac{1}{k} \quad \text{for } x = 1, 2, \ldots, k$$

and in this form it applies, for example, to the number of points we roll with a balanced die.

Remark 1.

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Exercise 2.

If X has a discrete uniform distribution $f(x) = \frac{1}{k}$ for $= 1, 2, \ldots, k$, show that

1. its mean is $\mu = \frac{k+1}{2}$;
2. its variance is $\sigma^2 = \frac{k^2-1}{12}$;
3. its *mgf* is given by

$$M_X(t) = \frac{e^t(1-e^{kt})}{k(1-e^t)}, \quad \text{for } t \neq 0.$$

Definition 2 (Bernoulli random variable).

A random variable X has a Bernoulli distribution and it is referred to as a Bernoulli random variable if and only if its probability distribution is given by

$$f(x;\theta) = \theta^x (1-\theta)^{1-x} \quad \text{for } x = 0, 1$$

Thus,

$$\mathbb{P}(X = 0) = 1 - \theta \quad \text{and} \quad \mathbb{P}(X = 1) = \theta,$$

for some $0 \leq \theta \leq 1$. We will often simply write $X \sim \mathrm{Ber}(\theta)$ to indicate such a random variable.

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Example 1.

Toss a coin once, corresponding to the sample space $\Omega = \{H, T\}$. Define X by $X(H) = 1$ and $X(T) = 0$. Then, X has a Bernoulli distribution.

Proposition 1 (Properties).

For a Bernoulli random variable.

- *The cdf is given by*

$$F_X(x;\theta) = \begin{cases} 0 & \textit{for } x < 0 \\ 1-\theta & \textit{for } 0 \le x < 1 \\ 1 & \textit{for } 1 \le x \end{cases}$$

- *The moment generating function is*

$$M_X(t) = (1-\theta) + \theta e^t.$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \theta \quad \textit{and} \quad \mathbb{V}ar(X) = \theta(1-\theta).$$

Definition 3 (Binomial random variable).

Consider n independent trials, each of which has a probability of success θ, and probability of failure $1-\theta$. If X represents the number of successes that occur in the n trials, then X is said to be a binomial random variable with parameters (n, θ). It is denoted as

$$X \sim \mathrm{Bin}(n, \theta)$$

- We will often simply write $X \sim \mathrm{Bin}(n, \theta)$ to indicate such a random variable.
- Note that the above implies that X only takes values in $\{0, 1, \dots, n\}$ with positive probability:

Example 2.

Toss the same coin n times. Let X be the number of heads witnessed in all n coin tosses. Assume that the probability of getting heads on each toss is $0 \leq \theta \leq 1$. Then, $X \sim \mathrm{Bin}(n, \theta)$.
To see this, consider the case in which the first k tosses result in heads (H) and the remainder result in tails (T). This is captured by the sample

$$\underbrace{HH \cdots H}_{k \text{ times}} \underbrace{TT \cdots T}_{n-k \text{ times}}.$$

This sample occurs with probability $\theta^k (1-\theta)^{n-k}$. However, there are $\binom{n}{k}$ permutations of the letters above, from which we obtain the expression

$$\mathbb{P}(\{X = k\}) = \binom{n}{k} \theta^k (1-\theta)^{n-k}.$$

Proposition 2 (Properties).

For a binomial random variable $X \sim \mathrm{Bin}(n, \theta)$*.*

- *The probability mass function is*

$$f(x; n, \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \quad \textit{for } x = 0, 1, 2, \cdots, n$$

- *Moment generating function:* $M_X(t) = \left(1 - \theta + \theta e^t\right)^n$*.*
- *Expectation and variance are:*

$$\mathbb{E}[X] = n\theta \quad \textit{and} \quad \mathbb{V}ar(X) = n\theta(1-\theta).$$

Proof.

- By the binomial theorem,

$$\sum_{x=0}^{n} f(x;n,\theta) = \sum_{x=0}^{n} \binom{n}{x} \theta^{x} (1-\theta)^{n-x} = (\theta + 1 - \theta)^{n} = 1^{n} = 1. \quad \square$$

- For the mgf, we have

$$\begin{aligned} M_X(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{x=0}^{n} e^{tx} f(x;n,\theta) &= \sum_{x=0}^{n} \binom{n}{x} \left(e^{t}\theta\right)^{x} (1-\theta)^{n-x} \\ &= \left(1 - \theta + \theta e^{t}\right)^{n}. \end{aligned}$$

Proof: Cont'd.

- Taking derivatives, $M'_X(t) = e^t n\theta M_X(t)^{(n-1)/n}$

$$M''_X(t) = M'_X(t) + e^{2t}\left(n-1\right) n\theta t^2 M_X(t)^{(n-2)/n}.$$

Therefore, $\mathbb{E}[X] = M'_X(0) = M_X(0)^{(n-1)/n} n\theta = n\theta$

$$\begin{aligned}\mathbb{E}\left[X^2\right] &= M''_X(0) = M'_X(0) + M_X(0)^{(n-2)/n}\left(n-1\right) n\theta^2 \\ &= n\theta\left(1 + \left(n-1\right)\theta\right)\end{aligned}$$

and hence

$$\mathrm{Var}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}X\right)^2 = n\theta\left(1 + \left(n-1\right)\theta\right) - \left(n\theta\right)^2 = n\theta\left(1-\theta\right).$$

$\square$

Example 3.

Find the probability of getting five heads and seven tails in 12 flips of a balanced coin.

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Solution:

- Let X be the number of heads, then $X \sim \text{Bin}\left(n = 12,\ \theta = \frac{1}{2}\right)$.
- Substituting $x = 5, n = 12$, and $\theta = \frac{1}{2}$ into the formula for the binomial distribution,
- we get

$$\mathbb{P}(X = 5) = \binom{12}{5}\left(\frac{1}{2}\right)^{5}\left(1 - \frac{1}{2}\right)^{12-5} \approx 0.19$$

Example 4.

Find the probability that 7 of 10 persons will recover from a tropical disease if we can assume independence and the probability is 0.80 that any one of them will recover from the disease.

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Solution:

- Let X be the number persons recover from a tropical disease, then $X \sim \text{Bin}\,(n = 10,\ \theta = 0.80)$.
- Substituting $x = 7, n = 10$, and $\theta = 0.80$ into the formula for the binomial distribution,
- we get

$$\mathbb{P}\,(X = 7) = \binom{10}{7}\,(0.80)^{5}\,(1 - 0.80)^{10-7} \approx 0.20$$

Proposition 3.

Let

$$X \sim \text{Bin}(n, \theta)$$
$$Y \sim \text{Bin}(n-1, \theta)$$

Then,

$$\mathbb{E}\left[X^k\right] = n\theta\mathbb{E}\left[(Y+1)^{k-1}\right]$$

Recall:

$$x\binom{n}{x} = n\binom{n-1}{x-1}. \tag{1}$$

Proof.

$$\mathbb{E}\left[X^k\right] = \sum_{x=0}^{n} x^k \binom{n}{x} \theta^x (1-\theta)^{n-x} = \sum_{x=1}^{n} x^k \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$\text{By } (1) = n\theta \sum_{x=1}^{n} x^{k-1} \binom{n-1}{x-1} \theta^{x-1} (1-\theta)^{n-1}.$$

Let $y = x - 1$. Therefore,

$$\mathbb{E}\left[X^k\right] = n\theta \sum_{y=0}^{n-1} (y+1)^{k-1} \underbrace{\binom{n-1}{y} \theta^y (1-\theta)^{n-1-y}}_{=\mathbb{P}(Y=y)}$$

$$= n\theta \sum_{y=0}^{n-1} (y+1)^{k-1} \mathbb{P}(Y=y) = n\theta \mathbb{E}\left[(Y+1)^{k-1}\right]$$

□

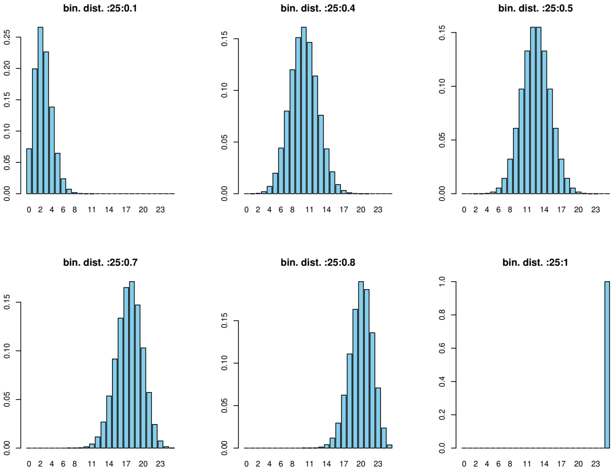

Figure 1: Illustration for Binomial distribution

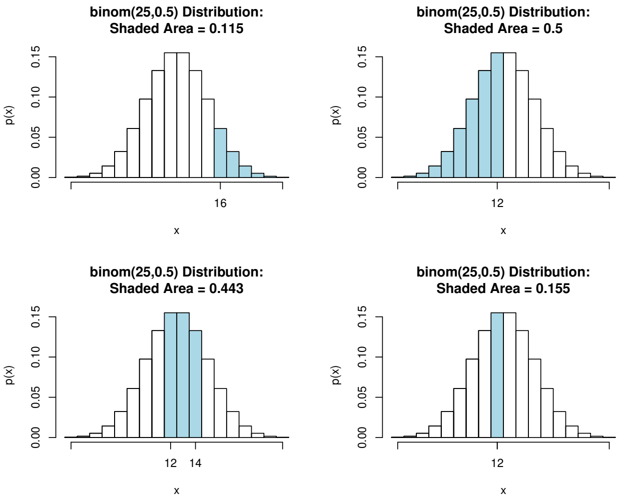

Figure 2: Calculating probabilities based on Binomial distribution

- In connection with repeated Bernoulli trials, we are sometimes interested in the number of the trial on which the k^{th} success occurs.
- For instance, we may be interested in the probability that:
  - the tenth child exposed to a contagious disease will be the third to catch it;
  - the probability that the fifth person to hear a rumor will be the first one to believe it;
  - or the probability that a burglar will be caught for the second time on his or her eighth job.

Definition 4 (Negative binomial random variable).

Suppose that independent trials, each having probability of success $0 < \theta < 1$, are performed until a total of r successes are accumulated. If X is the number of trials required, then X is said to be a negative binomial random variable.
It is denoted as

$$X \sim \mathrm{NB}(r, \theta)$$

The last trial must necessarily result in a success, and there must be $r-1$ more success in the first $x-1$ trials. Therefore, the probability distribution of X is

$$\begin{aligned} f(x; r, \theta) = \mathbb{P}(X = x) &= \binom{x-1}{r-1} \theta^{r-1} (1-\theta)^{x-r} \theta \\ &= \binom{x-1}{r-1} \theta^{r} (1-\theta)^{x-r} \end{aligned}$$

for $x = r, r+1, r+2, r+3, \ldots$

Example 5.

If the probability is 0.40 that a child exposed to a certain contagious disease will catch it, what is the probability that the tenth child exposed to the disease will be the third to catch it?

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Solution:

- Substituting $x = 10, r = 3$, and $\theta = 0.40$ into the formula for the negative binomial distribution, we get

$$f(10; 3, 0.40) = \binom{9}{2}(0.40)^3(0.60)^7 = 0.0645.$$

Recall the sum of a negative binomial series:

$$(1-x)^{-r} = \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} x^k \tag{2}$$

Proposition 4 (Properties).

For a Negative Binomial random variable $X \sim \mathrm{NB}(r, \theta)$*.*

- *Moment generating function:*

$$M_X(t) = \left(\frac{\theta e^t}{1 - (1-\theta)e^t} \right)^r \quad \textit{for} \quad t < -\log(1-\theta)$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \frac{r}{\theta} \quad \textit{and} \quad \mathbb{V}ar(X) = \frac{r(1-\theta)}{\theta^2}.$$

Proof.

For the mgf

$$
\begin{aligned}
M_X(t) = \mathbb{E}(e^{tX}) &= \sum_{x=r}^{\infty} e^{tx} \binom{x-1}{r-1} (1-\theta)^{x-r} \theta^r \times \frac{(e^t)^r}{(e^t)^r} \\
&= \theta^r (e^t)^r \sum_{x=r}^{\infty} e^{tx} \binom{x-1}{r-1} (1-\theta)^{x-r} (e^t)^{-r} \\
&= (\theta e^t)^r \sum_{x=r}^{\infty} \binom{x-1}{r-1} (1-\theta)^{x-r} (e^t)^{x-r} \\
&= \left(\theta e^t\right)^r \sum_{x=r}^{\infty} \binom{x-1}{r-1} [(1-\theta)e^t]^{x-r} \\
&= \left(\theta e^t\right)^r \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} \left[(1-\theta)e^t\right]^k, \text{put } k = x-r \\
&= (\theta e^t)^r \left[1-(1-\theta)e^t\right]^{-r} = \left(\frac{\theta e^t}{1-(1-\theta)e^t}\right)^r
\end{aligned}
$$

□

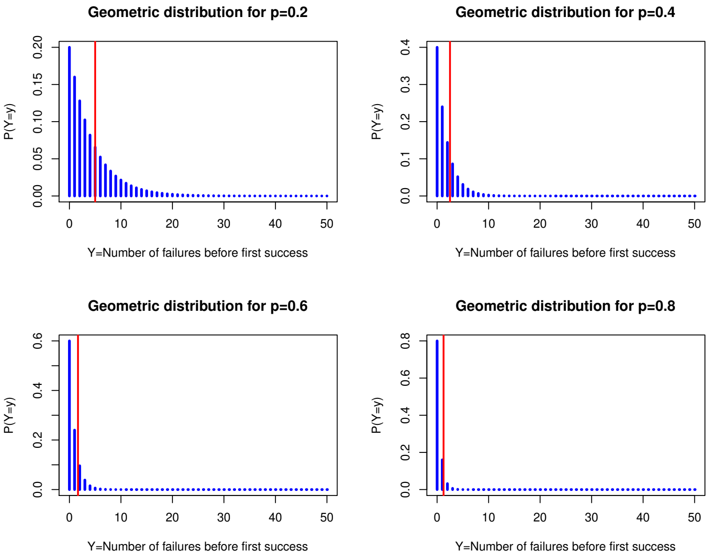
Definition 5 (Geometric random variable).

Suppose that independent trials, each having probability of success $0 < \theta < 1$, are performed until a success occurs. If X is the number of trials required, then X is said to have a geometric distribution. It is denoted as

$$X \sim \mathrm{Geo}(\theta)$$

The probability distribution if X is

$$f(x;\theta) = \mathbb{P}(X = x) = \theta(1-\theta)^{x-1} \quad \text{for } x = 1, 2, 3, \ldots$$


Geometric distribution for p=0.2
Geometric distribution for p=0.4
Geometric distribution for p=0.6
Geometric distribution for p=0.8
P(Y=y)
Y=Number of failures before first success
0.00 0.05 0.10 0.15 0.20
0.0 0.1 0.2 0.3 0.4
0.0 0.2 0.4 0.6
0.0 0.2 0.4 0.6 0.8
0 10 20 30 40 50

For a discrete random variable like the geometric distribution, the CDF is the sum of the probabilities for all outcomes up to x. Hence,

$$
\begin{aligned}
F_X(x) = P(X \leq x) &= \sum_{k=1}^{x} P(X = k) \\
&= \theta \sum_{k=0}^{x-1} (1-\theta)^k \\
&= \theta \frac{1-(1-\theta)^x}{1-(1-\theta)} \\
&= 1-(1-\theta)^x
\end{aligned}
$$

Proposition 5 (Properties).

For a Geometric random variable $X \sim \text{Geo}(\theta)$.

- *Moment generating function:*

$$M_X(t) = \frac{\theta e^t}{1 - (1-\theta)e^t} \quad \textit{for } t < -\ln(1-\theta).$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \frac{1}{\theta} \quad \textit{and} \quad \mathbb{V}ar(X) = \frac{(1-\theta)}{\theta^2}.$$

Proof.

- For the mgf,

$$M_X(t) = \sum_{k=1}^{\infty} e^{kt}\mathbb{P}(X=k) = \sum_{k=1}^{\infty} e^{kt}\theta(1-\theta)^{k-1}$$
$$= \theta e^t \sum_{k=1}^{\infty} e^{t(k-1)}(1-\theta)^{k-1}$$
$$= \frac{\theta e^t}{1-(1-\theta)e^t}$$

□

Proof: Cont'd.

Let $q = 1 - \theta$. Therefore,

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x q^{x-1}\theta = \sum_{x=1}^{\infty}(x-1+1)q^{x-1}\theta = \sum_{x=1}^{\infty}(x-1)q^{x-1}\theta + \sum_{x=1}^{\infty} q^{x-1}\theta$$

$$= \sum_{x=1}^{\infty}(x-1)q^{x-1}\theta + \frac{\theta}{1-q} = \sum_{x=1}^{\infty}(x-1)q^{x-1}\theta + \frac{\theta}{1-(1-\theta)}$$

$$= \sum_{x=1}^{\infty}(x-1)q^{x-1}\theta + 1 = \sum_{y=0}^{\infty} y q^{y}\theta + 1, \quad \text{put}, y = x-1$$

$$= q\sum_{y=0}^{\infty} y q^{y-1}\theta + 1 = q\mathbb{E}[X] + 1$$

$$\Longrightarrow \mathbb{E}[X](1-q) = 1 \Longrightarrow \mathbb{E}[X] = \frac{1}{1-q} = \frac{1}{\theta}.$$

□

Proof: End.

$$
\begin{aligned}
\mathbb{E}\left[X^2\right] &= \sum_{x=1}^{\infty} x^2 q^{x-1}\theta = \sum_{x=1}^{\infty}(x-1+1)^2 q^{x-1}\theta \\
&= \sum_{x=1}^{\infty}(x-1)^2 q^{x-1}\theta + \sum_{x=1}^{\infty} 2(x-1)q^{x-1}\theta + \sum_{x=1}^{\infty} q^{x-1}\theta \\
&= \sum_{y=0}^{\infty} y^2 q^y \theta + 2\sum_{y=1}^{\infty} y q^y \theta + 1 = q\mathbb{E}\left[X^2\right] + 2q\mathbb{E}[X] + 1
\end{aligned}
$$

$$\implies \theta\mathbb{E}\left[X^2\right] = \frac{2q}{\theta} + 1$$

$$\implies \mathbb{E}\left[X^2\right] = \frac{2q+\theta}{\theta^2} = \frac{2-\theta}{\theta^2}$$

Therefore,

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2 = \frac{2-\theta}{\theta^2} - \frac{1}{\theta^2} = \frac{1-\theta}{\theta^2}$$

$\square$

Example 6.

I am new to basketball and so I am practicing to shoot the ball through the basket. If the probability of success at each throw is 0.2, how many times would I fail on average until a success occurs.

Solution:

- Let X denote the number of shoots before I get a success.
- Now I assume that the individual throws are independent of each other, which might not be exactly true in real life.
- Under that assumption, $X \sim \mathrm{Geo}(0.2)$.
- So the average number of times until a success occurs is

$$\mathbb{E}(X) = \frac{1}{0.2} = 5.$$

Theorem 1.

Let i, j *be positive integers.* $X \sim \text{Geo}(\theta)$. *Then*

$$P(X \geq i + j | X \geq i) = P(X \geq j)$$

Proof.

First we notice that for $x > 0$,

$$P(X \geq x) = 1 - P(X \leq x - 1) = 1 - \left(1 - (1 - \theta)^{x-1}\right) = (1 - \theta)^{x}$$

Note that the formula holds for $x = 0$ also.

$$\begin{aligned} P(X \geq i + j | X \geq i) &= \frac{P(\{X \geq i + j\} \cap \{X \geq i\})}{P(X \geq i)} \\ &= \frac{P(X \geq i + j)}{P(X \geq i)} = \frac{(1 - \theta)^{(i+j-1)}}{(1 - \theta)^{i-1}} \\ &= (1 - \theta)^{j} = P(X \geq j) \end{aligned}$$

□

- To obtain a formula analogous to that of the binomial distribution that applies to sampling without replacement, in which case the trials are not independent,
- let us consider a set of N elements of which M are looked upon as successes and the other $N - M$ as failures.
- As in connection with the binomial distribution, we are interested in the probability of getting x successes in n trials, but now we are choosing, without replacement, n of the N elements contained in the set.

- There are $\binom{M}{x}$ ways of choosing x of the M successes
- and $\binom{N-M}{n-x}$ ways of choosing $n-x$ of the $N-M$ failures,
- and, hence, $\binom{M}{x}\binom{N-M}{n-x}$ ways of choosing x successes and $n-x$ failures.
- Since there are $\binom{N}{n}$ ways of choosing n of the N elements in the set, and we shall assume that they are all equally likely (which is what we mean when we say that the selection is random),
- the probability of "x successes in n trials" is

$$\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

Definition 6.

A random variable X has a hypergeometric distribution and it is referred to as a hypergeometric random variable if and only if its probability distribution is given by

$$f(x; n, N, M) = \mathbb{P}(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}},$$

for $x = 0, 1, 2, \ldots, n, \quad x \leq M$ and $n - x \leq N - M$

Example 7.

As part of an air-pollution survey, an inspector decides to examine the exhaust of 6 of a company's 24 trucks. If 4 of the company's trucks emit excessive amounts of pollutants, what is the probability that none of them will be included in the inspector's sample?

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Solution:

- Substituting $x = 0, n = 6, N = 24$, and $M = 4$ into the formula for the hypergeometric distribution, we get

$$f(0; 6, 24, 4) = \mathbb{P}(X = 0) = \frac{\binom{4}{0}\binom{20}{6}}{\binom{24}{6}} = 0.2880$$

Theorem 2.

The mean and the variance of the hypergeometric distribution are

$$\mu = \frac{nM}{N} \quad \text{and} \quad \sigma^2 = \frac{nM(N-M)(N-n)}{N^2(N-1)}$$

Exercise 3.

Among the 120 applicants for a job, only 80 are actually qualified. If 5 of the applicants are randomly selected for an in-depth interview, find the probability that only 2 of the 5 will be qualified for the job by using

(a) the formula for the hypergeometric distribution;

(b) the formula for the binomial distribution with $\theta = \frac{80}{120}$ as an approximation.

Solution 3.

1. Substituting $x = 2, n = 5, N = 120$, and $M = 80$ into the formula for the hypergeometric distribution, we get

$$f\,(2; 5, 120, 80) = \mathbb{P}\,(X = 2) = \frac{\binom{80}{2}\binom{40}{3}}{\binom{120}{5}} = 0.164$$

2. Substituting $x = 2, n = 5$, and $\theta = \frac{80}{120} = \frac{2}{3}$ into the formula for the binomial distribution, we get

$$f\left(2; 5, \frac{2}{3}\right) = \mathbb{P}\,(X = 2) = \binom{5}{2}\left(\frac{2}{3}\right)^2\left(1 - \frac{2}{3}\right)^3 = 0.165$$

As can be seen from these results, the approximation is very close.

Situation: Let the discrete random variable X denote the number of times an event occurs in an interval of time (or space). Then X may be a **Poisson** random variable with $x = 0, 1, 2, \ldots$

Examples

1. Let X equal the number of typos on a printed page. (This is an example of an interval of space - the space being the printed page.)
2. Let X equal the number of cars passing through the intersection of University and Campbell Avenue in one minute. (This is an example of an interval of time - the time being one minute.)
3. Let X equal the number of students arriving during office hours.

Definition 7.

A random variable X has a Poisson distribution with parameter $\lambda > 0$ if

$$\mathbb{P}\left(X = x\right) = \frac{\lambda^x}{x!}e^{-\lambda},$$

for $x = 0, 1, 2, \dots$ and $\lambda > 0$, where λ will be shown later to be both the mean and the variance of X. We will often simply write $X \sim \text{Poisson}(\lambda)$ to indicate such a random variable.

Proposition 6.

Let $X \sim \text{Poisson}(\lambda)$*. Then,*

$$\sum_{x \geq 0} (X = x) = 1.$$

Proof.

By the Taylor expansion of e^{λ},

$$\sum_{x \geq 0} (X = x)) = \sum_{x \geq 0} \frac{\lambda^x}{x!} e^{-\lambda} = e^{-\lambda} \sum_{x \geq 0} \frac{\lambda^x}{x!} = e^{-\lambda} e^{\lambda} = 1. \qquad \square$$

Proposition 7 (Properties).

For a Poisson random variable $X \sim \text{Poisson}(\lambda)$.

- *Moment generating function:*

$$M_X(t) = e^{\lambda\left(e^t - 1\right)}$$

- *Expectation and variance are:*

$$\mathbb{E}[X] = \lambda \quad \textit{and} \quad \mathbb{V}ar(X) = \lambda.$$

Proof.

the moment generating function:

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{x\geq 0} e^{tx}\frac{\lambda^x}{x!}e^{-\lambda} = e^{-\lambda}\sum_{x\geq 0}\frac{\left(\lambda e^t\right)^x}{x!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Taking derivatives,

$$M_X'(t) = \lambda e^t M_X(t), \quad M_X''(t) = M_X'(t)\left(\lambda e^t + 1\right)$$

Therefore,

$$\mathbb{E}[X] = M_X'(0) = \lambda e^0 M_X(0) = \lambda$$
$$\mathbb{E}\left[X^2\right] = M_X''(0) = M_X'(0)\left(\lambda e^0 + 1\right) = \lambda(\lambda+1)$$

and hence

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 = \lambda(\lambda+1) - \lambda^2 = \lambda.$$

□

One way to motivate the Poisson distribution is through the following observation:

Proposition 8.

Let $\lambda > 0$ *and suppose that* $n\theta \to \lambda$ *as* $n \to \infty$. *Then,*

$$\lim_{n\to\infty} \binom{n}{k} \theta^k (1-\theta)^{n-k} = \frac{\lambda^k}{k!} e^{-\lambda}, \qquad k = 0, 1, 2, \ldots$$

We recognize the left hand side in the above from $\mathrm{Bin}(n, \theta)$. The above suggests that $\mathrm{Poisson}(n\theta)$ captures the number of successes in n trials, each having probability θ, as the number of trials becomes large.

Example 8.

The number of market crashes per annum could be modeled as a $\mathrm{Poisson}(\lambda)$ random variable with, for example, $\lambda = 0.1$ (one crash every ten years).

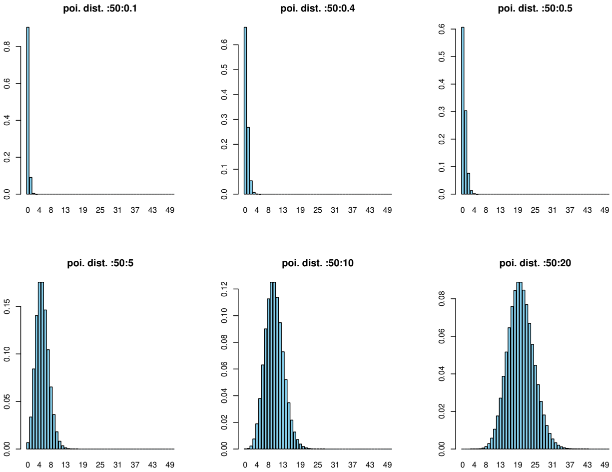

Figure 3: Illustration for Poisson distribution

Exercise 4.

Surveys indicated that 40% of the Zulu tribe members use the toothpaste 'Zebra'. A marketing company wants to interview a user of that toothpaste. Let X be the number of persons that the company would contact until a person who uses 'Zebra' will be found.

1. What is the distribution of X?
2. What is the expectation and variance of X?
3. What is the probability that the company will interview at most 3 persons until a 'Zebra' user is found?
4. Let Y be the number of clients that the company would contact until 2 persons who 'Zebra' will be found. What is the distribution of Y?
5. What is the expectation and variance of Y?

Solution 4.

1. X is a geometric random variable, with probability 0.4 for success. Therefore, $X \sim \text{Geo}(0.4)$
2. $X \sim \text{Geo}(0.4)$, then

$$\mathbb{E}[X] = \frac{1}{\theta} = \frac{1}{0.4} = 2.5$$
$$\mathbb{V}\text{ar}(X) = \frac{1-\theta}{\theta^2} = \frac{1-0.4}{(0.4)^2} = \frac{0.6}{0.16} = 3.75$$

3. $X \sim \text{Geo}(0.4)$, therefore,

$$\mathbb{P}(X \leq 3) = 1 - \mathbb{P}(X > 3) = 1 - \sum_{k=4}^{\infty} (1-\theta)^{k-1}\theta$$
$$= 1 - (1-p)^3 = 1 - (0.6)^3 = 1 - 0.216 = 0.784$$

4. $X \sim \text{NB}(2, 0.4)$
5. $X \sim \text{NB}(2, 0.4)$, therefore,

$$\mathbb{E}[X] = \frac{r}{\theta} = \frac{2}{0.4} = 5$$
$$\mathbb{V}\text{ar}(X) = \frac{r(1-\theta)}{\theta^2} = \frac{(2)(0.6)}{(0.4)^2} = \frac{0.12}{0.16} = 7.5$$

Exercise 5.

On average, 5.2 hurricanes hit a certain region in a year. What is the probability that there will be 3 or fewer hurricanes hitting this year?

Solution 5.

Let X be the number of hurricanes hitting the region in a year. Therefore,

$$X \sim \text{Poisson}(5.2)$$

Therefore,

$$\begin{aligned}
\mathbb{P}(X \leq 3) &= \sum_{i=0}^{3} \frac{e^{-\lambda}\lambda^i}{i!} \\
&= \sum_{i=0}^{3} \frac{e^{-5.2}(5.2)^i}{i!}
\end{aligned}$$

Exercise 6.

The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ. However, such a random variable can be observed only if it is positive, since if it is zero, then we cannot know that such an insect was on the leaf. Let Y denote the observed number of eggs, then

$$\mathbb{P}(Y = i) = \mathbb{P}(X = i \mid X > 0)$$

where X is a Poisson random variable with parameter λ. Find $\mathbb{E}[Y]$.

Solution 6.

$$
\begin{aligned}
\mathbb{E}[Y] &= \sum_{i=0}^{\infty} i\mathbb{P}(X = i \mid X > 0) \\
&= \sum_{i=0}^{\infty} i\frac{\mathbb{P}(X = i \cap X > 0)}{\mathbb{P}(X > 0)} \\
&= \sum_{i=1}^{\infty} i\frac{\mathbb{P}(X = i)}{\mathbb{P}(X > 0)} \\
&= \frac{\mathbb{P}[X]}{\mathbb{P}(X > 0)} \\
&= \frac{\lambda}{1 - e^{-\lambda}}
\end{aligned}
$$

- For your practice do the following exercises from text book: 5.10, 19, 20, 21, 23, 25, 29, 33, 41, 45, 47, 51, 57, 59, 63, 65, 69, 75, 77, 81

Main textbook:

- John E. Freund's. *Mathematical Statistics With Applications* Ed: 8 (Required).

Suggested textbooks:

- Ross, S., 2009. *A first course in probability*. Upper Saddle River, 6.
- Hossein Pishro-Nik. *Probability, Statistics and Random Processes* here.

Some useful websites:

- Society of Actuaries, Exam of probability, sample questions.
- Society of Actuaries, Exam of probability, sample solutions.